

Soil Sorption Estimator (SSE)

Requirements:

- SSE is a Streamlit-based web predictive platform that only requires a web-browser.
- Requires **internet connection** for deployment or online hosting.

GUI of the SSE Prediction Platform:

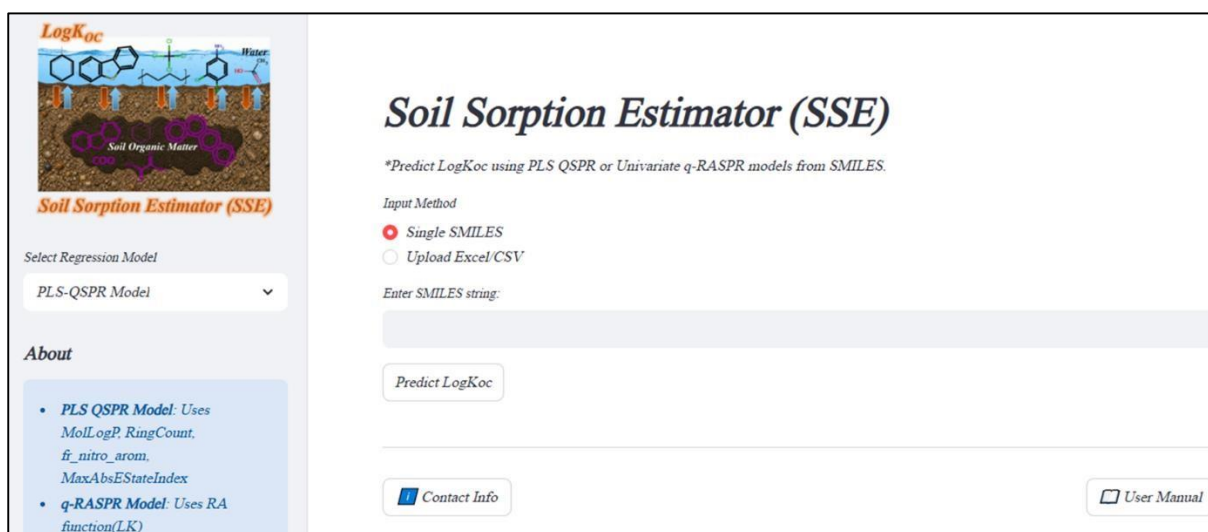


Figure 1: Soil Sorption Estimator GUI

***Note:-** The user needs a valid SMILES string for the compound that is to be predicted.

Prediction Procedure:

1. To predict a single compound, the user can simply use **Single SMILES** as **Input**
Method enter the SMILES string of the desired compound as input in the “Enter SMILES string” tab and then click the “Predict LogK_{OC}” button.
2. To predict a set of compounds, the user should prepare an Excel file (.xlsx format) containing the SMILES for each chemical in the first column of the first sheet with a proper heading. Then, the user must select “**Upload Excel**” as the input method and select the prepared file as input in the “Upload” tab by clicking on the button shown in **Figure 2**.
2. Afterward, click on the “Predict LogK_{OC}” button to obtain the predictions.

Figure 2: Steps to Upload SMILES file (.xlsx) for prediction

Input File:

- Must be in .xlsx format.
- A suitable heading in the first cell of the first column (**A1**).
- Valid SMILES in the first column from **Row 2**.

Table 1: Sample input file for the compounds needed to be predicted

	A	B	C
1	SMILES		
2	ccccc		
3	C(F)CCNC		
.....			
.....			

*The user can also refer to the “Sample.xlsx” file in the ‘data’ folder to prepare their input file.

View and Download Result:

Once the prediction is complete, the results can be viewed directly within the interface. The output will include details such as SMILES (for the input compounds), predicted LogK_{OC} values, and Applicability Domain (AD) status. Users also have the option to download the results in .csv format for single/batch process, which can be used for further analysis, recordkeeping, or reporting purposes.

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Tool developed by



(GUI developed by **Shubham Kumar Pandey and Souvik Pore**)

For further queries, please contact

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